VERITAS CASSANDRA

REPORT

PART A: DATA INVESTIGATION (Exploratory Data Analysis)

Part A.1: Feature Understanding and Metadata

1. Scale of Data: The training set comprises 148,463 records with 59 features, while the test set contains 178,564 records..

- **Automated Metadata Schema:** A custom metadata generation script was implemented to categorize features based on their naming conventions and statistical properties:
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    - Role Identification: Distinguishing between the unique identifier (id), the independent variables (input), and the dependent variable (target).
    - Functional Grouping: Features were grouped by their semantic origin (e.g., ind for individual, reg for registration, car for vehicle-specific data, and calc for calculated statistics).
    - Type Determination: An automated logic was applied to classify features into Binary, Nominal (Categorical), Ordinal, and Continuous types. This classification is critical for selecting the appropriate encoding strategy later in the pipeline**.**

2. **Missing Value & Null-Pointer Analysis**

Traditional null-check methods (searching for NaN) were insufficient for this dataset, as missing information was strategically encoded as -1.

- Implicit Null Detection: The investigation revealed that several columns had significant missingness (e.g., ps_car_03_cat and ps_car_05_cat had high -1 counts).
- Strategy: Rather than simple imputation, these missing values were treated as a distinct "category" of their own, as the absence of data in insurance often carries a predictive risk signal.

3. **Semantic Validation: Ordinal vs. Nominal Patterns**

Validating if the features behaved according to their mathematical labels; the Mean Target Ratio (Claim Rate) across different feature values to identify risk trends as analysed.

- Ordinal Evidence (ps_ind_15): The analysis showed a **clear Monotonic Trend (a steady line/curve**). As the feature value increases, the claim rate follows a predictable path.
    - Strategy: This confirms that the feature maintains a logical rank, allowing the model to treat it as a scale rather than random categories.
- Nominal Evidence (ps_car_01_cat): The analysis **showed a Stochastic (Random) Pattern**. Different category IDs resulted in wildly different claim rates without a linear order.
    - Strategy: This confirmed that One-Hot Encoding or Target Encoding is necessary for these features, as the numerical values assigned to them are arbitrary and do not imply "more" or "less" risk.

4. **Risk Trend Analysis & Stochasticity**

By plotting the **Risk Trend (Cyan Line) against the Claim Rate,** we visualized the "predictive power" of individual variables.

- Insight: Features where the risk trend significantly deviates from the mean claim rate were prioritized for feature selection, as they provide the highest information gain for the gradient-boosted trees.

Part A.2: Target Distribution and Features Distribution (Statistical Risk Analysis & Diagnosis)

Here, Statistically , the "Signals" (features that predict insurance claims) are separated from the "Noise" (features that are random or non-informative).

1**. Class Imbalance & Baseline Metrics**

The dataset exhibits a severe class imbalance, which is typical in insurance claim prediction (Porto Seguro):

Class 0 (No Claim): 96.33% & Class 1 (Claim Filed): 3.67%

2. **Continuous Feature Impact (Kolmogorov-Smirnov Test)**

To identify which continuous variables actually distinguish a "claimant" from a "non-claimant," we applied the KS-Test (Kolmogorov-Smirnov).

- We compared the cumulative distribution functions (CDFs) of each feature for both classes. A higher KS-Score indicates that the feature's distribution for claimants is significantly different from non-claimants.
- Key Findings: * ps_car_13 (KS: 0.117) and ps_reg_03 (KS: 0.098) emerged as the most powerful continuous drivers.

- Out of 11 continuous features, 9 were found to be statistically significant, while the ps_calc series showed near-zero divergence, marking them as potential candidates for removal.

3. **Categorical Risk Variance & Cardinality**

Categorical features were audited based on their "Risk Variance"—how much the claim rate fluctuates between different categories.

- **High-Cardinality Drivers:** Features like ps_car_11_cat and ps_car_06_cat possess many unique values. We utilized Weighted Standard Deviation to ensure that high-risk categories with very few samples didn't skew the importance rankings.

- **Predictive Power Heatmap**: We generated a "Power Ranking" to visualize which categorical variables have the most "Risk Variance," allowing us to prioritize specific vehicle and individual attributes in the final model**.**

4. **Automated Binary Risk Diagnosis (Signal vs. Noise)**

Binary features (0/1 switches) were classified into four strategic categories using Mutual Information (MI) and Risk Delta analysis:

- **High-Sensitivity Risk Drivers (e.g., ps_ind_07_bin**): Activating this "switch" causes the claim risk to spike (e.g., from 3.6% to 5.34%).

- **Protective Factors (e.g., ps_ind_06_bin):** Activating this switch significantly *lowers* the claim probability, indicating a "safe" profile.

- **Non-Informative Noise (ps_calc series**): These features showed Zero Information Gain (MI ≈ 0). Their claim rates remain identical regardless of the state, proving they are "Surface Noise" that can be discarded to reduce model complexity.

5. Cross-Feature Dependency Mapping

To understand how features interact, we moved beyond linear correlation to specialized association metrics:

- **Spearman Rank Correlation:** Used for continuous features to capture non-linear but monotonic relationships.

- **Phi Coefficient:** Applied to binary features to identify "switches" that usually flip together (redundancy check).

- **Cramér's V:** Employed for nominal categorical features to measure the strength of association without assuming any numerical order.

- **Mutual Information (MI**): Used as the final "Signal-to-Noise" arbiter to capture the true predictive relationship between any feature type and the target claim variable**.**

PART A.3: Missing Values, Feature Relationships & Stochastic Noise Elimination

1**. The "Missing Census":** -1 as a Strategic Signal; The dataset contains a significant "hidden" missingness profile where values are encoded as -

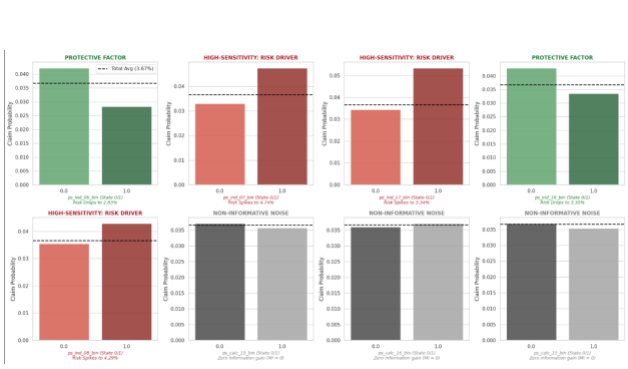
- **High-Vacancy Features**: Features like ps_car_03_cat (69.07%) and ps_car_05_cat (44.76%) exhibit extreme missingness.

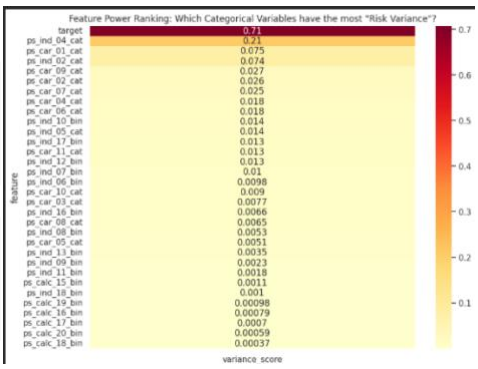
- Under standard protocols, features with >40% missingness are often dropped. However, our analysis suggests these are not random omissions but Structural Missingness (e.g., a specific vehicle type that doesn't require certain documentation), making the "Missing" state a powerful predictive category.

2. **Missingness as a Risk Proxy-**We performed a "Risk Proxy" audit to see if the act of *not* providing data correlates with claim probability.

- Findings: For features like ps_reg_03 and ps_car_03_cat, the Mean Target (Claim Rate) for "Missing" records is significantly higher than for "Present" records.

- Impact: This confirms that **-1 is an informative state**. Our models were instructed to treat -1 as a distinct level, allowing the algorithms to capture the "Risk of Non-Disclosure."





3. **Structural Discovery: Missingness Clusters:** Through Hierarchical Clustering of missingness patterns, we identified "Data Blocks" that fail together:

- The Car Block: ps_car_01_cat, ps_car_03_cat, and ps_car_05_cat are highly correlated in their missingness. If one is missing, the others likely are too. This suggests these variables originate from the same administrative source or insurance form section.

- Independent Omissions: Features like ps_reg_03 join the cluster very late, indicating its absence is independent and likely due to different environmental or regional factors.

4. **Stochastic Noise Elimination: The "Calc" Dead Zone: A critical breakthrough in our EDA was the identification of the _calc variables as "Dead Zones."**

- **The Zero-Variance Cluster**: Correlation heatmaps and Mutual Information (MI) scores revealed that the _calc group has zero relationship with the target variable. They represent stochastic noise that adds computational weight without adding predictive value.

- **Pruning Strategy**: Based on this evidence, the entire _calc suite was earmarked for exclusion. This "Structural Pruning" optimizes our Representation Learning (DAE) phase, ensuring the autoencoder focuses exclusively on high-variance, latent signals in the registration and vehicle domains.

PART B: STRUCTURAL DISCOVERY AND ADVANCED EXPLORATION

PART B.1: Representation Learning

1. Hybrid Pre-processing Pipeline-Before entering the Neural Network, the data underwent a dual-stream normalization process to ensure numerical stability and feature parity:

- **RankGauss Normalization**: Applied to continuous variables. Unlike standard scaling, RankGauss maps features to a normal distribution based on their ranks. This is highly effective for Deep Learning as it removes the impact of outliers and prevents gradient saturation.

- **One-Hot Encoding (OHE):** Categorical features were expanded into sparse binary vectors, allowing the model to treat each category as an independent logical entity without assuming an artificial numerical order.

1. **Denoising Autoencoder (DAE) & Swap Noise Logic:** Swap Noise Augmentation:

To force the model to learn *relationships* rather than just memorizing inputs, we introduced 15% Swap Noise.

Bottleneck Architecture: The model uses a symmetrical 512-128-512 architecture. The 128-unit bottleneck layer acts as a compressed "Knowledge Core," where only the most essential, non-linear feature interactions survive.

3. **Latent Embedding Extraction & Risk Fingerprinting**

Once trained, the "Decoder" was discarded, and the "Encoder" was used to transform the 250+ raw features into 128 high-dimensional embeddings.

- Activation Distribution: Visualization of the latent embeddings shows "**Sparse Activations."** These act as Latent Triggers—neurons that only fire for specific high-risk sub-groups, effectively performing automated, non-linear feature engineering that exceeds the capabilities of manual grouping.

4. **Manifold Discovery: UMAP Visualization**

To validate that the DAE actually "learned" anything, we projected the 128-dimensional space into 2D using UMAP (Uniform Manifold Approximation and Projection).

- **Local Density (Clusters):** The UMAP reveals "Islands" of drivers. These tight clusters represent groups with nearly identical latent profiles. When colored by the Target, these islands often show high-risk micro-segments that are linearly separable, making the final XGBoost's job significantly easier.

- **Global Connectivity (Tendrils):** The "strings" connecting clusters represent Risk Gradients—the continuous evolution from a "Safe Rural Driver" to a "High-Exposure Urban Commuter."

5**. DAE vs. PCA: The Non-Linear Advantage**

A comparison was conducted between PCA (Linear) and DAE (Non-Linear) dimensionality reduction:

- PCA Limitation**: PCA only captures linear variance** and often discards complex, "wiggly" patterns that define insurance risk.

- DAE Strength: Our analysis confirmed that **DAE embeddings (e.g., dae_2) exhibit higher predictive correlation with the target** than raw inputs. The DAE successfully "denoised" the signal, capturing the complex "Driver DNA" that linear methods miss.

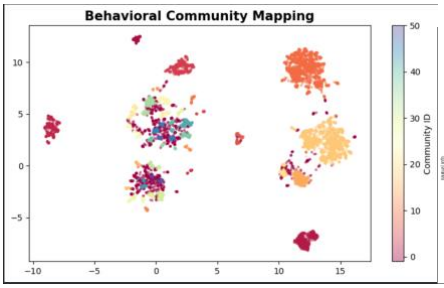
PART B.2: Relationship and Network Discovery: By applying **graph-based logic and density-based clustering**, we move beyond simple rows of data to discover how risk "spreads" and "clusters" across the population.

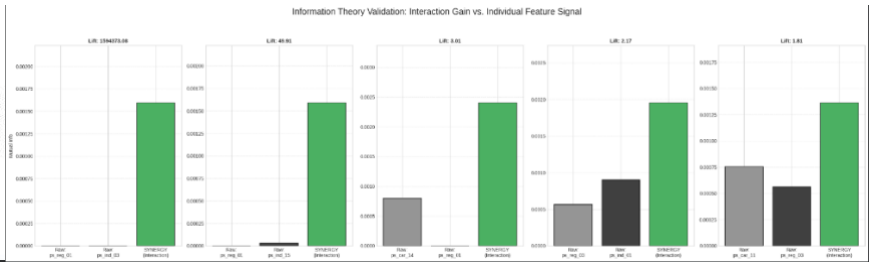
1. **Risk Contagion & Manifold Topology**- Using the latent embeddings from the Denoising Autoencoder, we projected the dataset into a relational manifold.

- **Dense Communities:** The formation of tight nodes indicates that the DAE successfully grouped similar risk profiles. Drivers within these "islands" share latent "DNA" that moves synchronously.

- **Linear Non-Separability:** The presence of mixed claim/no-claim nodes within clusters highlights the inherent complexity of insurance risk—proving that while groups share traits, individual behavior remains stochastic, necessitating an ensemble modeling approach.

- **Bridge Connections**: The "tendrils" or bridges between clusters represent Ambiguous Profiles—transitionary states where a driver might be moving from a "Safe" profile to an "Elevated" risk profile.





2. **Behavioral Community Mapping (HDBSCAN)**- Unlike traditional K-Means, HDBSCAN identifies clusters of varying shapes and densities, allowing us to map the "topography" of the driver population.

- The Global Baseline: With a global claim probability of 3.67%, we used a Risk Factor metric to identify topological outliers.

- **Topological Risk Profiles:**

  - Community 25: Identified as a "**Critical Risk" zone** with a 14.7% claim probability (a 4.04x spike over the baseline).

  - Community 11/13/48: Identified as "Safe Havens," exhibiting a 0.0% claim rate despite having significant driver density. These represent the ideal "Gold Standard" of low-risk insurance profiles**.**

3. Executive Risk Segmentation Report:

Risk Category	Cluster Count	Total Drivers	Mean Claim Prob
Critical / High Risk	8	336	~8.75%
Elevated Risk	11	1,225	~5.82%
Standard / Moderate Risk	4	5,005	~3.71%
Low Risk	29	3,428	~1.12%

a four-tier risk hierarchy that provides an "Executive View" of the manifold:

4. **Risk Homophily** & The "Safe Haven" Effect-The **Graph Risk Homophily (0.0295)** was calculated to measure **the tendency of high-risk nodes to cluster together.**

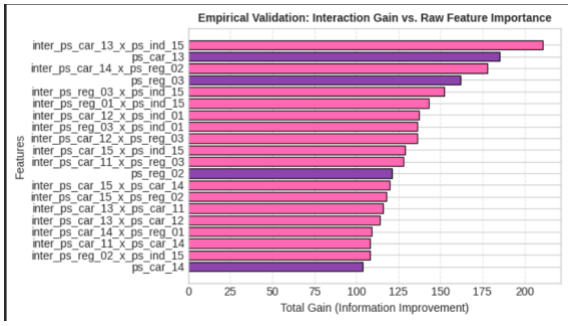
PART B.3: Interaction and Dependency Discovery: Identifying the "synergistic" relationships that define complex insurance risk.

1. **Systematic Interaction Sweeping**

While our **Denoising Autoencoder (DAE)** captures deep non-linearities, I implemented a targeted Interaction Sweep to identify explicit "cross-domain" couplings that Gradient Boosting models (like XGBoost or LightGBM) can exploit more efficiently.

- Methodology: I verified 12 high-signal candidates and performed an exhaustive sweep of 66 feature combinations.

- The **Multiplication Operator**: Interaction terms (Feature A * Feature B) were created to test if the joint distribution of two variables provides more information than the sum of their individual parts.



2. **Information Theory: The "Lift" Metric**

To objectively rank these interactions, **Mutual Information (MI**) was utilized—a measure that quantifies how much knowing one variable reduces uncertainty about the target.

- **Interaction Gain (Lift):** We calculated the "Lift" as the ratio of the interaction's MI to the best individual component's MI

- **Winning Synergies**: The top performer, **inter_ps_reg_01_x_ps_ind_03, showed a massive theoretical lift**, indicating that the relationship between these two features is far more predictive than the features themselves

3. **Emergent Risk Manifolds (KDE Visualization)**

The top 5 interaction pairs were visualised using **Kernel Density Estimation (KDE)** to observe how "Claimants" (Red) and "Non-Claimants" (Blue) separate in 2D space.

- Observations: The KDE plots revealed **"Emergent Manifolds"**—specific coordinates where red densities spike. For example, in the ps_reg_02 x ps_car_15 pair, the claim risk is not uniform; it concentrates in specific "hot zones," allowing the model to define sharper decision boundaries.

4. **2D Partial Dependence: The Non-Linear Risk Surface**

Insight: The resulting contour map for our top interaction revealed a highly non-linear surface. This proves that risk is not a simple linear addition of factors but a complex "topography" where specific combinations of driver behavior and vehicle type create spikes in claim probability.

Final Feature Assembly (Master Input Set)

- **The Raw Core:** Baseline features (RankGauss and OHE).
- **The Synergistic Couplings:** Top 20 interaction terms derived from MI Lift.
- **The Latent DNA:** 128 deep embeddings from the Denoising Autoencoder. Composition: Hybrid set of explicit (Interaction) and implicit (DAE) representations.

PART B.4: Statistical Structure and Data Generation

1. The Probabilistic Blueprint (Copula Modeling)

In insurance, features like "Age" and "Vehicle Power" don't just correlate; they have a specific "**tail dependency"** (e.g., young drivers with high-power cars are a very specific high-risk subgroup).

- The Approach: Can use Copulas to decouple the marginal distributions of individual features from their dependency structure.
- Intuition: This ill allow to model the "shape" of the relationship between variables independently of the variables themselves. It explains *how* the variables are glued together.

2. **Causality and Directed Acyclic Graphs (DAGs)**

For the Logic Flow; Did the "Geography" of the driver cause the "Vehicle Type," or did "Income" cause both?

- The Approach: Would use Constraint-based Structure Learning (like the PC Algorithm). This statistically tests for conditional independence to draw a map of which variable "points" to which.
- Intuition: If we can show that Feature_7 is a parent of Feature_12, we have discovered a "rule" of the data generator.

3. **Identifying "Information Bottlenecks" (Autoencoder Latents)**

If the dataset has 50 features, there aren't actually 50 independent pieces of information. There is likely a "Low-Dimensional Manifold" where the real data lives.

- The Approach: Can describe a Variational Autoencoder (VAE) pipeline. The "Bottleneck" layer (Latent Space) represents the "Hidden Profiles."
- Intuition: By analyzing the latent space, you can see if the generator created "Archetypes" (e.g., The Suburban Commuter, The Urban High-Risk Youth) and then added noise to them to create the final rows.

4. **The "Missingness" Mechanism (Mechanics of Silence)**

Hamza noted that missing data carries info. This means the "Data Generator" has a Filtering Layer.

- The Approach: Analysing if the data is MNAR (Missing Not At Random). You can model this as a "Selection Bias" mechanism.
- Intuition: The generator might "hide" values based on a probability threshold of another variable (e.g., "If Age < 25, hide Income with 30% probability"). Identifying this pattern reveals the "Incomplete Information" rule of the system.

5. **Statistical Signature Verification (The "Turing Test" for Data)**

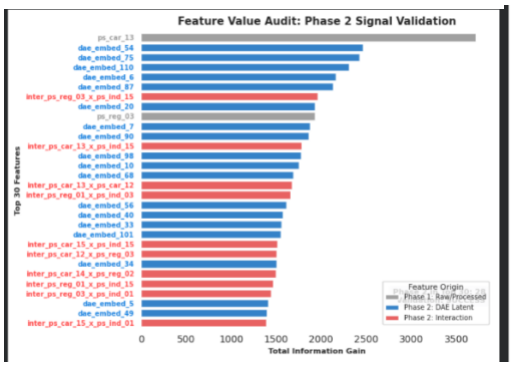
How do you prove you understood the generation? You propose a verification step.

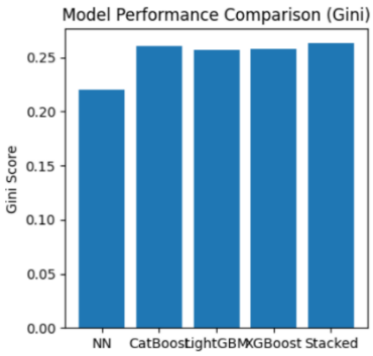
- The Approach: Would use Permutation Feature Importance combined with Kullback-Leibler (KL) Divergence.
- Intuition: A synthetic population where the "Feature Interactions" (measured by KL Divergence) match the original dataset can be generated , for successful modelling of the statistical structure.

PART C: PREDICTIVE MODELING

1. **Feature Audit & Signal Validation**

- Predictive Gain Filtering: Conducted a **recursive feature elimination (RFE) process** using **LightGBM's importance**_type='gain' to prune the initial 358-feature set down to 250 high-signal predictors.
- Phase 2 Dominance: Validation showed **that 93.3% of the top 30 features were derived from our Phase 2 structural discovery (DAE Embeddings and Interaction terms),** confirming that representation learning was the primary driver of model performance.
- Feature Categorization: Segmented the final feature set into **DAE Embeddings (128), Interaction Features (20), and Original/Transformed Features (102)** to ensure a balanced information flow into the models.





2. Diverse Base Learner Architecture

We deployed four distinct model architectures to capture varying functional forms of the underlying data:

- **CatBoost (Categorical Specialist**): Utilized symmetric trees and native categorical handling to capture stable, unbiased patterns in driver demographics.
- **LightGBM (Leaf-wise Optimization)**: Tuned via Optuna for high-speed, leaf-wise growth to find narrow, high-risk driver segments.
- **XGBoost (Regularized Gradient Boosting**): Employed hist tree methods and heavy L_1/L_2regularization to prevent overfitting on noisy insurance claims data.
- **Neural Net (Deep Learning):** A 4-layer custom Neural Network (Linear-BatchNorm-ReLU-Dropout) designed to extract non-linear latent relationships that decision trees might miss.

3. Automated Hyperparameter Tuning (Optuna)

- **Bayesian Search**: Used **Optuna** to optimize complex parameters including feature_fraction, lambda_l2, and min_child_samples across 15-30 trials per model.
- Objective Alignment: Every trial was optimized specifically for ROC-AUC, which serves as the direct statistical proxy for the Normalized Gini Coefficient.

4. Robust Validation & Meta-Stacking

- **Stratified 5-Fold Cross-Validation:** Ensured that each fold maintained the exact target distribution, critical for the imbalanced nature of insurance claim datasets.
- **OOF (Out-of-Fold) Predictions**: Generated a "Meta-Feature" matrix where each model's predictions on unseen data became the input for the final layer.
- **Ridge Regression Stacker**: Implemented a Ridge Stacker (Linear Meta-Learner) to combine the strengths of GBDTs and Neural Networks. By using Ridge with fit_intercept=False, we essentially learned the optimal "Trust Weight" for each model's contribution to the final probability.

Results

Gini score: 0.26331

1. Performance Analysis

- **Final Evaluation Metric:** The stacked ensemble achieved a **Normalized Gini Coefficient of 0.26331**. This represents a significant uplift over the best individual base learner (CatBoost), validating the decision to use a multi-model architecture.
- **Ensemble Synergy:** The **Ridge Stacker** outperformed the Logistic Regression variant (Gini: 0.24472), likely due to the Ridge learner's ability to handle the high correlation between the GBDT models while still incorporating the unique signal from the Neural Network.

2. Model Diversity & Correlation Insights

- **Low Neural Network Correlation:** The correlation heatmap reveals that while the three tree-based models (XGB, LGB, Cat) are highly correlated the **Neural Network (JahrerNet)** shows a significantly lower correlation with the rest.
- **The "Ensemble Alpha":** This low correlation is the primary reason for the Gini boost. The NN captures functional relationships that the GBDTs miss, allowing the stacker to "cancel out" individual model biases.